CCNA

NETWORKING FUNDAMENTALS

Study Guide | Theory, Examples and Practical Understanding

INTRODUCTION TO NETWORKING | NETWORK DEVICES | OSI & TCP/IP MODELS

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1. Introduction to Networking

A computer network is a collection of two or more computing devices connected so they can communicate and share data, resources, and services. Devices may connect through copper cables, fiber-optic cables, or wireless signals.

1.1 What Is a Computer Network?

A computer network is a group of interconnected devices that exchange information using agreed communication rules called protocols.

For example, computers in an office may connect to the same network to share files, use a network printer, access an internal server, or reach the internet. Communication requires network interfaces, transmission media, addressing, network devices, and protocols.

Component	Purpose	Examples
End devices / hosts	Create, send and receive information	PCs, laptops, servers, printers, smartphones
Network interface	Connects a device to a network	Ethernet NIC, Wi-Fi adapter
Transmission medium	Carries signals between devices	Copper cable, fiber, radio waves
Network devices	Connect, forward or control traffic	Switch, router, access point, firewall
Protocols	Define communication rules and formats	Ethernet, IP, TCP, UDP, DNS

Example Ten office computers connect to an Ethernet switch. They can communicate with a file server and network printer. A router can connect this local network to other networks and the internet.

1.2 Importance of Networking

- Resource sharing: Multiple users can share printers, storage, applications, and internet connections.
- Communication: Networks support email, messaging, voice calls, and video conferencing.
- Centralized management: User accounts, policies, software, and backups can be managed centrally.
- Data sharing: Authorized users can exchange files and information efficiently.
- Scalability: New devices, users, and locations can be added as an organization grows.
- Cost efficiency: Shared infrastructure can reduce equipment and operating costs.
- Remote access: Users can securely access organizational resources from other locations.

Networking and Cybersecurity

Greater connectivity also creates security risks. Unauthorized access, malware propagation, data interception, denial-of-service attacks, and misconfiguration can affect network availability,

confidentiality, and integrity. Segmentation, firewalls, encryption, access control, patching, and monitoring help reduce these risks.

1.3 Types of Networks

Networks are commonly classified by coverage area, purpose, or connection technology. A network may belong to more than one category.

Type	Meaning	Typical use
PAN	Personal Area Network	A short-range network around one person, such as a phone connected to Bluetooth earbuds or a smartwatch.
LAN	Local Area Network	Connects devices in a limited area such as a home, lab, office, or building.
WLAN	Wireless Local Area Network	A LAN that uses wireless technology, usually Wi-Fi, to connect devices.
MAN	Metropolitan Area Network	Connects networks across a city or metropolitan area.
WAN	Wide Area Network	Connects networks over large geographic areas, such as multiple cities or countries.

WLAN is not a separate geographic-size category in the same way as LAN or WAN. It describes a LAN that uses wireless access. The internet is a global interconnection of networks.

1.4 Network Topologies

A network topology describes how devices and links are arranged. Physical topology describes the actual cabling and device connections; logical topology describes how data moves through the network.

Topology	Structure	Advantages and limitations
Bus	All devices share one backbone cable.	Can be simple and inexpensive, but a backbone failure can disrupt the network; troubleshooting and shared-medium performance can be difficult.
Star	Each device connects to a central hub or switch.	Easy to manage; a single endpoint cable failure usually affects only that endpoint.

		Failure of the central device can affect many devices.
Ring	Each device connects to its two neighboring devices, forming a loop.	Can provide an orderly path for traffic; the effect of a link failure depends on the ring design and redundancy.
Mesh	Devices have multiple interconnections.	Offers redundancy and resilience, but requires more links, ports, cost, and configuration.
Hybrid	Combines two or more topology types.	Flexible for real-world requirements, but can be more complex to design and troubleshoot.

Practical note Modern Ethernet LANs commonly use star or extended-star designs. End devices connect to access switches, and switches connect to other switches or routers through uplinks.

1.5 Peer-to-Peer vs. Client-Server Network

These are two common network architectures. In a peer-to-peer (P2P) network, devices can share resources directly. In a client-server network, a server provides services and clients request those services.

Aspect	Peer-to-Peer	Client-Server
Structure	Devices can act as both clients and resource providers.	Dedicated servers provide services to client devices.
Management	Usually distributed; devices are managed individually.	Centralized management is possible.
Cost	Can be inexpensive for small environments.	May require server hardware, software, and administration.
Security and backup	Consistent policies and backups can be harder to maintain.	Centralized authentication, policies, and backups are easier to apply.
Typical use	Small home or informal file-sharing environments.	Businesses, schools, and larger organizations.

Examples P2P: Two laptops share files directly. Client-server: A laptop authenticates to a file server and accesses a shared folder.

1.6 Internet, Intranet and Extranet

Term	Definition	Typical users
Internet	The global public interconnection of networks and services.	The public, organizations, and service providers.
Intranet	An organization's private internal network or internal web resources.	Authorized employees or members.
Extranet	A controlled part of an organization's resources made available to selected external parties.	The organization and approved partners, suppliers, or customers.

Security note: A private intranet is not automatically secure. Authentication, authorization, segmentation, encryption, and logging are still necessary. Extranet users should receive only the access required for their work.

2. Network Devices

Network devices are hardware or virtual components that connect systems, forward traffic, repeat signals, or enforce network policies. Understanding each device's function and the information it uses to make decisions is essential for networking.

2.1 Router

A router connects different IP networks and forwards packets toward their destinations. It examines the destination IP address and uses a routing table to select an appropriate route and next hop.

- A traditional router primarily operates at OSI Layer 3 (Network Layer).
- It forwards packets between different IP networks or subnets.
- It can use static routes or learn routes through dynamic routing protocols.
- Many home routers also provide NAT, DHCP, wireless access, and basic firewall features; these are additional functions, not the core definition of routing.

Example When a home computer sends a packet to an internet server, it sends the packet to its default gateway. The router forwards it toward the next network.

2.2 Switch

A Layer 2 Ethernet switch connects devices within a LAN or VLAN and forwards Ethernet frames. It learns source MAC addresses and builds a MAC address table. It uses the destination MAC address to decide which port should receive a frame.

- A standard Layer 2 switch operates at the Data Link Layer.
- Each switch port creates a separate collision domain.
- If the destination MAC address is unknown, or the frame is a broadcast, the switch floods it to relevant ports in the VLAN, excluding the port on which it arrived.
- A Layer 3 switch can also perform IP routing.

2.3 Hub

A hub is a legacy Layer 1 device. It repeats an incoming electrical signal out through its other ports. It does not inspect destination MAC addresses or selectively forward Ethernet frames. Devices connected to a hub share bandwidth and can experience collisions. Switches have largely replaced hubs in modern Ethernet LANs.

2.4 Bridge

A bridge connects LAN segments and uses MAC addresses to filter or forward frames. It operates at the Data Link Layer. The bridge concept is important for understanding switching: an Ethernet switch can be viewed as a high-speed multiport bridge.

2.5 Repeater

A repeater operates at the Physical Layer. It receives and regenerates a signal to help extend the distance over which it can travel. It does not filter traffic based on MAC or IP addresses.

2.6 Wireless Access Point (WAP)

A wireless access point allows Wi-Fi clients to connect to a wired LAN. It manages wireless radio communication and may advertise a network name called an SSID. An access point commonly bridges wireless clients to the wired network.

- SSID is the name used to identify a wireless network.
- WPA2 or WPA3 should be used to protect wireless access, with strong credentials and secure configuration.
- A wireless access point is not necessarily a wireless router. A wireless router commonly combines routing, switching, and access-point functions.

2.7 Modem

The word modem comes from modulator-demodulator. A modem terminates or converts signals according to an ISP access technology. Cable, DSL, and cellular modems are used with different access networks. Fiber connections commonly use an Optical Network Terminal (ONT); it is not always accurate to call an ONT a modem.

2.8 Firewall

A firewall allows or blocks network traffic according to a security policy. Depending on its type, it can inspect source and destination IP addresses, protocols, ports, connection state, and application-level information.

- A network firewall can control traffic between network segments or at a network perimeter.
- A host-based firewall controls incoming and outgoing traffic for an individual computer or server.
- A firewall is only one security control. Secure configuration, updates, monitoring, and other defenses are also necessary.

2.9 Network Interface Card (NIC)

A Network Interface Card (NIC) is the hardware or virtual interface that allows a device to connect to a network. An Ethernet NIC uses a wired connection; a wireless NIC uses radio communication such as Wi-Fi. Link speed, duplex mode, and the interface's MAC address are important networking properties.

2.10 Server and Client

A server is a hardware or software system that provides services or resources to other systems. A client is a device or application that requests those services. The same computer can act as a client in one situation and a server in another.

Server type	Purpose	Example
Web server	Delivers websites or web applications.	HTTP/HTTPS service
File server	Provides centralized file storage and sharing.	Shared folders
DNS server	Resolves domain names to IP	A domain-name lookup

	addresses.	
DHCP server	Leases IP configuration to clients.	IP address, subnet mask, gateway, DNS
Mail server	Supports sending, receiving, or storing email.	SMTP and IMAP services

2.11 Gateway

A gateway is a point through which traffic enters or leaves a network or system. In IP networking, a host's default gateway is usually the router interface used as the next hop when the destination is outside the local subnet. In other contexts, a gateway may also translate between protocols or application systems.

Do not confuse these terms A default gateway is the next-hop address configured on a host. A router is a network device. In a typical home network, the router's LAN interface acts as the default gateway.

2.12 Quick Comparison of Network Devices

Device	Typical OSI layer	Main function
Hub	Layer 1	Repeats signals; does not inspect addresses.
Repeater	Layer 1	Regenerates signals to extend transmission distance.
Bridge	Layer 2	Forwards or filters frames using MAC addresses.
Switch	Layer 2; Layer 3 for multilayer switches	Uses a MAC table to forward frames; Layer 3 switches can route IP packets.
Router	Layer 3	Uses destination IP addresses and a routing table to forward packets.
Firewall	Multiple layers, depending on type	Inspects and controls traffic according to security policy.
Access point	Primarily Layer 2 bridging, with Layer 1 radio	Connects wireless clients to a LAN.

3. OSI Model and TCP/IP Model

Network communication involves many operations. Layered models divide these operations into manageable functions, making network design, learning, and troubleshooting easier. The OSI model is a reference model; TCP/IP describes the protocol architecture used by the internet.

3.1 What Is the OSI Model?

The Open Systems Interconnection (OSI) model is a seven-layer conceptual reference model developed by the International Organization for Standardization (ISO). Each layer describes a group of networking functions and provides services to the layer above it. Real-world protocols do not always fit perfectly into one layer, but the model is very useful for learning and troubleshooting.

Layer	Name	Main responsibility	Examples / concepts
7	Application	Provides network services used by applications.	HTTP, DNS, SMTP
6	Presentation	Describes data representation, encoding, compression, and encryption.	Text encoding, serialization, encryption concepts
5	Session	Describes the establishment, management, and termination of communication sessions.	Session and dialog management
4	Transport	Provides process-to-process delivery, ports, segmentation, and transport reliability.	TCP, UDP
3	Network	Provides logical addressing and packet forwarding between networks.	IPv4, IPv6, routing, ICMP
2	Data Link	Provides local-link framing, MAC addressing, and media access functions.	Ethernet, Wi-Fi MAC, switching
1	Physical	Transmits bits as signals over a physical medium.	Copper, fiber, radio, connectors

Layer order From top to bottom: Application → Presentation → Session → Transport → Network → Data Link → Physical.

Layer-by-Layer Explanation

Layer 7 – Application: Provides network services to application processes. Examples include a browser using HTTP or an application making a DNS query.

Layer 6 – Presentation: Describes how data is represented so that systems can interpret it. Data formatting, encoding, compression, and encryption are commonly discussed at this layer. In real networks, technologies such as TLS may span or interact with multiple layers.

Layer 5 – Session: Describes how communication sessions are established, maintained, and terminated. In TCP/IP networks, these responsibilities are not always implemented as a separate layer; applications and libraries often handle them.

Layer 4 – Transport: Supports communication between application processes and uses port numbers to identify services. TCP provides a reliable, connection-oriented byte stream. UDP provides a connectionless datagram service.

Layer 3 – Network: Uses logical IP addresses and routes packets between networks. Routers make forwarding decisions primarily at this layer.

Layer 2 – Data Link: Handles local-link framing, MAC addressing, and media-access functions. Ethernet switches commonly forward frames using destination MAC addresses.

Layer 1 – Physical: Transmits bits as electrical, optical, or radio signals. Cables, connectors, radio frequencies, and signal characteristics are Physical Layer topics.

3.2 TCP/IP Model

The TCP/IP model is a practical layered model associated with the protocol suite used for internet communication. In the common four-layer version, some OSI layers are combined.

TCP/IP layer	Responsibilities	Approximate OSI mapping	Examples
Application	Application services, data representation, and session-related functions.	OSI Layers 5–7	HTTP, HTTPS, DNS, DHCP, SMTP, SSH
Transport	Process-to-process delivery, ports, and transport services.	OSI Layer 4	TCP, UDP
Internet	IP addressing and packet routing.	OSI Layer 3	IPv4, IPv6, ICMP
Network Access / Link	Local network framing and physical	OSI Layers 1–2	Ethernet, Wi-Fi

	transmission.		

Some learning materials use a five-layer TCP/IP model by separating the Physical and Data Link functions. The number of layers can vary by reference; focus on the functions and protocol roles.

OSI vs. TCP/IP

Aspect	OSI Model	TCP/IP Model
Type	Conceptual reference model.	Protocol architecture associated with internet communication.
Layers	Seven layers.	Commonly four layers; sometimes represented as five.
Main use	Learning, describing functions, and troubleshooting.	Understanding practical internet protocol architecture.
Layer mapping	More granular separation of functions.	Combines some OSI layers.

3.3 Encapsulation and Decapsulation

Encapsulation occurs when data moves down the sender's protocol stack and each layer adds control information, usually in a header and sometimes a trailer. Decapsulation is the reverse process: the receiver processes and removes the relevant protocol information as data moves up the stack.

A simplified example of a web request:

1. Application: The browser creates an HTTP request.
2. Transport: TCP adds its header, including source and destination port information. The result is a TCP segment.
3. Internet: IP adds source and destination IP addresses. The result is an IP packet.
4. Network Access: Ethernet adds a frame header and trailer. The result is an Ethernet frame.
5. Physical: The frame is transmitted as bits represented by signals over the cable or wireless medium.

At the receiving system, the process is reversed. The receiver processes the frame, packet, segment, and application data in sequence.

Important The Layer 2 frame can change at every routed hop because each link needs a frame addressed to the next-hop device. The source and destination IP addresses generally remain end-to-end, except when mechanisms such as Network Address Translation (NAT) modify them.

3.4 Protocol Data Units (PDU)

A Protocol Data Unit (PDU) is the name given to the data unit handled at a particular protocol layer. The name changes as data moves through the networking stack.

Layer / stage	PDU name	Description
Application / upper layers	Data	Information produced by an application.
Transport (TCP)	Segment	Application data with a TCP header. UDP data is commonly called a datagram.
Network / Internet	Packet	Data with an IP header containing source and destination IP addresses.
Data Link	Frame	Data with link-layer headers and trailers; Ethernet frames contain MAC addresses.
Physical	Bits	Binary values transmitted as physical signals.

Terminology note: A TCP segment and a UDP datagram are not the same thing. Some introductory diagrams use the word segment generically for the Transport Layer, but protocol-specific terminology is more accurate.

3.5 How Data Travels Across a Network

Imagine that a laptop is used to open a website. The following is a simplified overview; actual communication may also involve DNS caches, TLS, proxies, NAT, multiple routing decisions, and server-side processing.

6. The user enters a website's domain name in a browser.
7. If necessary, the device uses a DNS resolver to find the IP address associated with the domain name.
8. The application begins communication with the server. For HTTPS, TLS is used to establish a secure communication channel.
9. The transport layer may use TCP. Some applications use UDP-based transport; for example, HTTP/3 uses QUIC over UDP.
10. The IP layer creates packets using source and destination IP addresses. If the destination is outside the local subnet, the host sends the packet to its default gateway.
11. The link layer creates a local frame. The destination MAC address is normally the MAC address of the next hop, not the MAC address of the remote web server.
12. The switch forwards the frame across the local LAN. The router processes the IP packet, consults its routing table, selects the next hop, and creates a new Layer 2 frame for the next link.
13. The packet travels through one or more routers and networks until it reaches the destination server. The server processes the request and sends a response, which may return along a different route.

Key concept IP addresses identify endpoints across IP networks, while MAC addresses support delivery on a local link. A router removes the incoming Layer 2 frame information and forwards the

packet in a new frame for the next link.

Troubleshooting with the Layer Model

Symptom	Possible layer or issue	Useful checks / commands
No link light or connection	Physical: cable, port, NIC, wireless signal.	Check cable, port status, adapter status; Linux: `ip link`.
Devices on the same LAN cannot communicate	Data Link: VLAN, MAC learning, Wi-Fi association.	Check switch port/VLAN, wireless association, `arp -a`.
Local network works but external networks do not	Network: IP address, subnet mask, default gateway, routing.	Windows: `ipconfig`; Linux: `ip addr`, `ip route`; test with `ping`.
IP connectivity works but domain names fail	Application: DNS resolution.	`nslookup`, `dig`.
Server is reachable but a particular service fails	Transport/Application: port, firewall, service availability.	Windows: `Test-NetConnection`; Linux: `ss -tuln`.

4. Review Questions

1. What is a computer network?

Answer: A connection of two or more devices that enables communication and resource sharing.

2. What is the main difference between LAN and WAN?

Answer: A LAN covers a limited local area; a WAN connects networks across larger geographic areas.

3. Is WLAN different from LAN?

Answer: WLAN is a LAN that uses wireless technology, commonly Wi-Fi.

4. What is the role of the central device in a star topology?

Answer: All endpoint devices connect through it. If the central device fails, many or all connections may be affected.

5. How are a switch and a router different?

Answer: A typical Layer 2 switch forwards frames using MAC addresses within a LAN or VLAN. A router forwards IP packets between networks using a routing table.

6. Why is a hub less efficient than a switch?

Answer: A hub repeats signals to all other ports and does not selectively forward frames. A switch can forward frames to the appropriate port using its MAC address table.

7. What is a default gateway?

Answer: The next-hop address a host uses to reach destinations outside its local subnet, usually a router interface.

8. Which OSI layer is associated with routing?

Answer: Layer 3, the Network Layer.

9. At which OSI layer do TCP and UDP operate?

Answer: Layer 4, the Transport Layer.

10. What is encapsulation?

Answer: The process of adding protocol control information as data moves down the sender's networking stack.

11. What is the difference between a packet and a frame?

Answer: A packet is the Network Layer PDU; a frame is the Data Link Layer PDU.

12. Why do MAC addresses change across a router hop?

Answer: Each local link requires a new Layer 2 frame addressed to the next-hop device.

5. Glossary

Term	Definition
Bandwidth	The capacity of a link to carry data over a period of time.
Latency	The time delay for data to travel from one point

	to another.
Protocol	An agreed set of rules for communication.
Host	An end device or system connected to a network.
MAC address	A link-layer address used to identify a network interface on a local network.
IP address	A logical address used to identify an interface on an IP network.
Subnet	A logical division of an IP network.
Default gateway	The default next hop a host uses to reach other networks.
Port number	A Transport Layer number used to identify an application process or service.
VLAN	A logical segmentation of a Layer 2 network into separate broadcast domains.
PDU	The data unit handled by a particular protocol layer.
Encapsulation	The process of adding protocol information as data moves through network layers.